Discussion 1 OS Development

Discussion Topic:

The changes from early computers with no operating system to the sophisticated operating systems of today have been remarkable. In your initial discussion post, provide your thoughts on the following three questions:

1. What do you think is the most revolutionary development in operating systems over the past 50 years?
2. How did the development contribute to the technology available to us today?
3. What makes the development more revolutionary than other substantial developments?

My Post:

Hello Class

To answer the question about the most revolutionary development in operating systems (OSs) over the past 50 years, in my opinion, is the development of time-sharing systems, notably the one implemented within OSs like Unix/Linux, Windows, macOS, Android, and other cloud and mobile operating systems. Before the advent of OSs, programmers were required to work with machine language, punch cards, and manual job control. This made the computing process slow and difficult. OSs significantly improved the computing process by managing memory, processes, software, hardware, and user interaction in a more organized way (CSU Global, n.d.).

Time-sharing allowed computers to be transformed from single-task systems into systems that could share resources among multiple users and processes. Denning (2015) argues that OSs have evolved from batch processing to interactive, distributed, and cloud-mobile computing. These OS's developments also include process scheduling and other processes, such as memory management, file and storage management, device management, input/output operations, security management, and loading and execution. Process scheduling is the process by which the OS selects a process from a ready queue to run on the CPU. In other words, the process scheduler decides which process or job gets CPU time and how to balance performance, response time, fairness, and resource use (González-Rodríguez et al., 2024).

Time-sharing was a revolutionary OS development, and with the help of other OS processes, such as memory management, file and storage management, device management, input/output operations, security management, and loading and execution, it contributed directly to the technology used today for daily computing. For example, without resource management that utilizes multitasking and time-sharing, it would be difficult to run a browser, stream music, receive notifications, update files, and run security processes at the same time. Additionally, these advancements made possible by time-sharing in collaboration with other OS processes make modern mobile and cloud systems possible. For example, Android relies on a Linux-based kernel, and Linux has become important in mobile devices, embedded systems, cloud computing, and high-performance computing (Android, 2026; Perlow, 2020). This same idea of sharing resources while keeping processes separated is also found in containerization, where containers are isolated environments that allow applications and their dependencies to run separately from other applications. These containers provide OS-level virtualization by using OS-level isolation to separate processes and control access to CPU, memory, and storage resources (IBM, n.d.).

-Alex

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